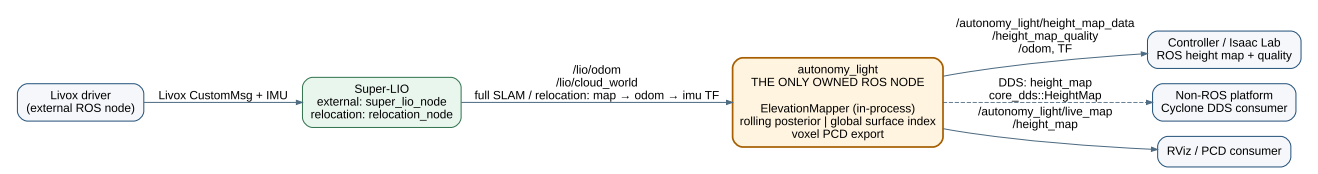


autonomy_light software architecture

Super-LIO integration and elevation mapping · 2026-08-03

Design decision. SLAM and relocalization are Super-LIO responsibilities. This package owns one ROS node only: it converts Super-LIO output into a robust elevation map and a sparse PCD.



Node responsibilities

Node	Owner	Responsibility
Livox driver	external	Publishes one Livox CustomMsg and IMU stream.
Super-LIO	external	LiDAR–IMU odometry, Scan Context + GICP + iSAM2 full SLAM, saved-map relocation, and continuous map → odom → imu TF.
autonomy_light	this package	One node: base pose/TF, rolling or global elevation map, quality output, live PCD and PCD save.

Data and TF contract

Direction	Topic	Meaning
input	/lio/odom	Super-LIO LiDAR pose; converted to base_link.
input	/lio/cloud_world	Registered cloud in the current global frame.
output	/autonomy_light/height_map_data	Controller distance grid.
output	/autonomy_light/height_map_quality	Per-cell validity, variance, age.
output	/autonomy_light/live_map	Sparse voxel PCD for RViz and map saving.
optional output	height_map	Direct Cyclone DDS core_dds::HeightMap payload for a non-ROS peer.

TF is odom|map → base_link → base_link_gravity and static base_link → lidar_link. Normal operation uses odom; a saved map runs Super-LIO relocation_node and uses map. After its NDT → ICP registration succeeds, Super-LIO continuously publishes map → odom → imu; the

transform is only recalculated at the configured low global-registration rate. Mapping runs the pose graph in Super-LIO. There is no `world` frame or duplicate SLAM backend in this package.

Elevation source selection

Configuration	Algorithm	Use
<code>height_map.source:rolling</code>	Bounded Bayesian ground/upper posterior with measurement variance, aging and outlier gate.	Default for control and sim-to-real.
<code>height_map.source:global</code>	Persistent, PCD-derived sparse surface index.	Repeatable terrain reference from a saved or accumulated map.

Both sources use the same output message. `valid=0` means the distance value is an unknown placeholder, not measured flat ground; Isaac Lab should consume the quality channels.

Transport selection

`height_map_output.transport` selects `ros2`, `cyclone_dds`, or `both`. The direct DDS option uses the installed IDL `core_dds::HeightMap { sequence<float> data; }`, configurable domain and topic, and best-effort keep-last delivery. Its flat, row-major data sequence has exactly the same values and unknown convention as ROS `HeightMap.data`; geometry and quality remain ROS-message metadata. `both` is the migration/validation mode.

Removed ownership

- Duplicate local pose graph, Scan Context loop closure, and map reconstruction
- FPFH/GICP initial localization and runtime map correction outside Super-LIO
- Secondary LiDAR merger and custom ROS-domain bridge executors
- Separate raw/refined/ROI map pipelines and duplicated path subscription

These removals make the state path linear: Super-LIO localizes; one node maps elevation.